

TEZPUR UNIVERSITY
Autumn Semester End Examination, 2021
CO309: OPERATING SYSTEM

TU/CSE

Full Marks: 60

Time: 3 hrs

All questions are compulsory. Numbers in right hand side indicate marks. Write the answers precisely.

1. Answer the following suitably (any four)

[2x4=8]

- a. Difference between mutex and binary semaphore
- b. Relation between working set model and page fault frequency
- c. Difference between LFU and MFU Algorithm
- d. "First-fit and best-fit are better than worst-fit in terms of speed and storage utilization". Justify the statement
- e. How do you implement a sleeplock?
- f. Differentiate between anticipatory paging and prepaging

2. What is the principle of optimality during page replacement services? What do you understand by the statement "more frames equal to more fault"?

[1+2=3]

3. In what situation the Dining Philosophers lock may lead to the possibility of deadlock? How does monitor based implementation helps to overcome from such a situation?

[2+2=4]

4. State the role of *readcount* and *wrt* semaphore in reader process of first readers-writers lock. (Note: writing the code is not asked)

[4]

5. If there are 64 frames available for allocation of 5 processes with size 1, 6, 3, 2 and 4 KB respectively in a virtual memory session, calculate the number of frames allocated for each of the processes for their allocation. Consider the proportional allocation policy is adopted by the memory management module.

[4]

6. What is locality in the context of working set model? Suppose there are 3 processes P1, P2, P3 having their locality size window to be 8, 20 and 5, respectively. If there are 19 frames available to be used prove that even on suspending process P1 the snapshot will lead to thrashing situation.

[2+3=5]

7. "Waiting on the condition variable automatically unlocks the mutex". Justify the statement for monitor tool implementation.

[4]

8. Consider a machine implements a virtual address space (VAS) having page size of 8KB. If it uses a 2-level paging scheme with 128 total entries in outer page table, where each page is of size 4KB, Calculate the size of the VAS scheme of the machine's OS.

[4]

9. Consider a 32-bit machine that uses main memory with four page frames and the following sequence of page references: 3,8,2,3,9,1,6,3,8,9,3,6,2,1,3. If it wants to use the page replacement policies First In First Out (FIFO) and Least Recently Used (LRU) which one will be preferable and why? (Note: you need to draw the allocation diagram for frames)

[4]

10. What is the role of *valid/invalid bit* used in page table implementation? Given a memory model requires to maintain 25% of page hit ratio. If the effective access time and memory access time of the model is 5 μ s and 1 μ s respectively what will be the TLB look up time of the model?

[1+3=4]

11. How do you define link and symbolic link in file management? What is a device directory? How does acyclic graph help in implementing file structure?

[2+1+2=5]

12. What are the models used to implement file access protection system? State the differences between them. Discuss role of elevator algorithm used for disk scheduling in I/O management.

[1+2+2=5]

13. Write notes on (any three) (Note: Answer in maximum three/four sentences only)

[2x3=6]

- a. Inverted page table
- b. Enhanced second chance algorithm
- c. I-node
- d. Log-structured File System (LFS)
- e. Solid State Drive (SSD)

$$32 \text{ bits} = 4 \text{ B}$$

$$2^2 = 4$$

=====

$$128 = 2^7$$

$$2^7 = 128$$

$$2^7 = 128$$

$$2^7 = 128$$

$$2^7 = 128$$

$$2^7 = 128$$

$$128$$

$$4 \text{ KB}$$

$$128 \text{ byte}$$

$$2^{12}$$

$$(32)$$

$$2 \times 2 = 4$$

$$2 \times 2 = 4$$

$$2 \times 2 = 4$$

$$2^5 = 32$$

$$2^5 = 32$$

$$2^5 = 32$$

$$2^5 = 32$$

$$2^5 = 32$$

$$2^5 = 32$$

$$2^5 = 32$$

$$2^5 = 32$$

$$2^5 = 32$$

$$2^5 = 32$$